

MR. AWAIS QURESHI
DOCTORAL RESEARCHER @ NUST SEecs

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Location: Pakistan (Willing to Relocate)

PROFESSIONAL SUMMARY

Applied AI researcher and software engineering professional with 14+ years of experience across academia and industry. Currently pursuing a PhD in Information Technology with specialized focus in Deep Learning, conducting research in longitudinal microbiome modeling, disease prediction, and AI-driven healthcare applications.

Experienced in teaching and mentoring students in Artificial Intelligence, Machine Learning, Data Structures, Programming, and Software Engineering. Skilled in designing and implementing deep learning models using TensorFlow and Python, while leveraging extensive software development expertise in C#, .NET, and cloud technologies. Passionate about combining research excellence, innovative teaching practices, and industry experience to prepare students for real-world challenges.

RESEARCH EXPERIENCE

PhD Researcher (Deep Learning & Genomics) | NUST-SEecs | 2019 – Present

- **Dynabiome Framework (Longitudinal Analysis):** Designed an interpretable unsupervised learning framework using LSTMs and Transformers to detect dysbiosis in temporal gut microbiome data. Solved data sparsity issues using CLR transformations and zero-inflation handling.
- **MetaResNet (CNN Architecture):** Architected a novel Convolutional Neural Network (CNN) for disease classification, optimizing colormap representations of sparse metagenomic datasets to improve F1-scores on imbalanced classes.
- **HCT Dysbiosis Prediction:** Developed an interpretable AI model to predict complications in Hematopoietic Cell Transplantation (HCT) patients, translating complex "black box" model decisions into clinically relevant biomarkers.
- **Tech Stack:** Python, TensorFlow, PyTorch, Keras, Pandas, Scikit-learn, Google Colab.

PROFESSIONAL EXPERIENCE

KOHAT UNIVERSITY OF SCIENCE AND TECHNOLOGY (KUST) | OCT 2020 – PRESENT

- **HEAD OF THE DEPARTMENT (JUL 2024 – OCT 2025):** Appointed to lead academic operations, curriculum development, and departmental accreditation compliance.
- **SENIOR LECTURER/FYP COORDINATOR (OCT 2020 – PRESENT):** Delivering graduate-level CS Curriculum and Coordinating Technical research projects.

TECHNICAL TEAM LEAD (REMOTE/PART-TIME) | BITSYM PVT LTD | JUN 2022 – May 2026

- Led development of subscription-based e-learning platform using Moodle.
- Integrated local payment gateways (EasyPaisa, JazzCash, MeezanPay).
- Delivered ERP and Odoo-based solutions for international clients.

SENIOR SOFTWARE DEVELOPER | SIPARADIGM DIAGNOSTIC INFORMATICS | JUN 2019 – NOV 2020

- Developed medical billing and claims management system using ASP.NET MVC.

- Improved patient data management and automated claims processing.

SENIOR SOFTWARE DEVELOPER | BITSYM PVT LTD | AUG 2018 – JUN 2019

- Built SDN network management system using PHP Laravel.
- Enhanced automation and network monitoring capabilities.

SELECTED PUBLICATIONS & MANUSCRIPTS

- **Qureshi, A.** *MetaResNet: Enhancing Microbiome-Based Disease Classification through Colormap Optimization and Imbalanced Learning.*
(Submitted, BMC Bioinformatics, 2026)
- **Qureshi, A.** *DynaBiome: Interpretable Unsupervised Learning of Gut Microbiome Dysbiosis using Temporal Deep Models.*
(PUBLISHED, BMC Bioinformatics, 2026)
- **Qureshi, A.** *DynaBiomeX: An Interpretable Dual-Strategy Deep Learning Framework for Architectural Noise Filtration in Sparse Longitudinal Microbiome Data.*
(Submitted: BMC Medical Informatics and Decision Making)

EDUCATION

PHD IN INFORMATION TECHNOLOGY | NUST - SEecs, ISLAMABAD (THESIS SUBMITTED)

- **Thesis :** Deep Temporal Models for Longitudinal Microbiome Analysis.

MS IN INFORMATION TECHNOLOGY (2015) NUST-SEecs, ISLAMABAD: 2015

- **Thesis:** Scalability Analysis of Genome Sequencing with MapReduce in Cloud.
- **Key Achievement:** Awarded AWS Research Credits to deploy and benchmark three Hadoop Clusters on AWS EC2, analyzing genomic data pipeline scalability.

MASTER OF COMPUTER SCIENCE (MCS), (KUST): 2006

TECHNICAL SKILLS

- **Deep Learning & AI:** TensorFlow, PyTorch, Keras, CNN, LSTM, Transformers, Scikit-learn.
- **Big Data & Cloud:** Apache Spark, Hadoop, AWS (EC2, S3).
- **Programming Languages:** Python (Advanced), SQL, C#, Java, R (Basic).
- **DevOps & Tools:** Docker, Git/GitHub, Linux/Unix Shell.
- **Web & Systems:** ASP.NET MVC, Laravel, RESTful APIs.

SPECIALIZATIONS

1. BUILDING GENAI APPLICATIONS AND AGENTS (COURSERA - 2026)
2. AWS GENERATIVE AI AND AI AGENTS WITH AMAZON BEDROCK (AWS - 2026)

CERTIFICATIONS

1. TENSORFLOW FOR AI/ML – DEEPLARNING.AI - 2022
2. CONVOLUTIONAL NEURAL NETWORKS – DEEPLARNING.AI - 2022
3. BIG DATA & HADOOP – UC SAN DIEGO, COURSERA- 2016